

## VOLGOGRAD GUMRAK, Russian Federation

WMO: 345600

Lat: 48.783N

Lon: 44.346E

Elev: 482

StdP: 14.44

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.3       | -3.1 | -13.7                         | 2.7 | -8.0 | -7.9 | 3.7 | -2.4 | 30.0                  | 24.9 | 27.5 | 23.4 | 9.5                   | 70   | 0.641 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 96.4            | 65.9 | 92.9 | 65.6 | 89.5 | 65.0 | 69.4                | 85.7 | 68.1 | 84.6 | 66.9 | 83.2 | 11.7                 | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.7                            | 93.1 | 73.4 | 62.9 | 87.3 | 72.8 | 61.2 | 82.1 | 72.3 | 33.7          | 85.9 | 32.7 | 84.2 | 31.7 | 82.8 | 76.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean  |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min   | Max   |                    | Min                                                | Max       | Min   | Max        | Min   | Max        | Min   | Max        |       |
| (a)               | (b)  | (c)  |                            | (e)   | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 26.9              | 23.8 | 21.4 | DB                         | -11.5 | 100.0 | 8.1                | 3.4                                                | -17.3     | 102.4 | -22.1      | 104.4 | -26.6      | 106.3 | -32.5      | 108.8 |
|                   |      |      | WB                         | -11.8 | 72.9  | 7.9                | 1.7                                                | -17.5     | 74.1  | -22.2      | 75.2  | -26.6      | 76.1  | -32.3      | 77.4  |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 47.9  | 20.0  | 21.2  | 32.6  | 49.6  | 61.9  | 70.6  | 75.4  | 73.8  | 61.4  | 47.6  | 33.9  | 24.5  | (6)  |
| (7)  |                                                                   | DBStd       | 21.95 | 11.74 | 11.76 | 9.10  | 8.17  | 8.18  | 7.62  | 6.40  | 7.45  | 8.20  | 8.94  | 9.54  | 10.63 | (7)  |
| (8)  |                                                                   | HDD50       | 3830  | 931   | 807   | 541   | 106   | 9     | 0     | 0     | 9     | 153   | 484   | 790   |       | (8)  |
| (9)  |                                                                   | HDD65       | 7189  | 1396  | 1227  | 1005  | 465   | 159   | 31    | 3     | 14    | 160   | 542   | 932   | 1255  | (9)  |
| (10) |                                                                   | CDD50       | 3051  | 0     | 0     | 1     | 93    | 378   | 618   | 788   | 738   | 353   | 79    | 3     | 0     | (10) |
| (11) |                                                                   | CDD65       | 932   | 0     | 0     | 0     | 1     | 64    | 199   | 326   | 286   | 54    | 2     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 10345 | 0     | 0     | 0     | 24    | 679   | 2203  | 3650  | 3280  | 493   | 16    | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 4510  | 0     | 0     | 0     | 1     | 201   | 940   | 1698  | 1537  | 130   | 3     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 10.9  | 11.9  | 12.6  | 12.6  | 11.5  | 10.3  | 9.7   | 9.2   | 9.4   | 9.9   | 10.3  | 11.0  | 12.0  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 14.9  | 1.3   | 1.1   | 1.1   | 1.0   | 1.7   | 1.7   | 1.1   | 0.9   | 1.2   | 1.2   | 1.1   | 1.5   | (15) |
| (16) |                                                                   | PrecMax     | 20.8  | 2.6   | 2.4   | 2.2   | 3.4   | 3.7   | 4.7   | 3.0   | 2.6   | 4.1   | 3.9   | 3.1   | 3.7   | (16) |
| (17) |                                                                   | PrecMin     | 9.9   | 0.6   | 0.6   | 0.2   | 0.1   | 0.4   | 0.1   | 0.1   | 0.0   | 0.2   | 0.3   | 0.4   |       | (17) |
| (18) |                                                                   | PrecStd     | 3.1   | 0.6   | 0.4   | 0.6   | 0.8   | 1.0   | 1.3   | 0.7   | 0.7   | 1.1   | 0.9   | 0.7   | 0.7   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 43.8  | 44.6  | 59.3  | 77.3  | 89.9  | 97.2  | 100.3 | 100.3 | 89.4  | 75.2  | 58.2  | 46.3  | (19) |
| (20) |                                                                   | MCWB        | 41.3  | 41.1  | 48.2  | 58.4  | 62.8  | 65.8  | 66.3  | 66.6  | 63.9  | 59.1  | 50.7  | 44.9  |       | (20) |
| (21) |                                                                   | 2%          | DB    | 39.1  | 39.6  | 53.6  | 72.0  | 85.6  | 93.0  | 96.0  | 96.4  | 83.7  | 69.5  | 53.3  | 43.0  | (21) |
| (22) |                                                                   | MCWB        | 38.3  | 37.6  | 44.8  | 56.1  | 62.3  | 65.6  | 66.4  | 65.7  | 61.1  | 56.2  | 48.6  | 41.9  |       | (22) |
| (23) |                                                                   | 5%          | DB    | 35.7  | 37.0  | 49.2  | 67.8  | 81.1  | 89.3  | 92.9  | 92.1  | 79.7  | 64.7  | 49.7  | 40.4  | (23) |
| (24) |                                                                   | MCWB        | 34.8  | 35.7  | 42.7  | 53.7  | 61.2  | 65.2  | 66.1  | 65.1  | 59.4  | 54.2  | 47.1  | 39.4  |       | (24) |
| (25) |                                                                   | 10%         | DB    | 33.8  | 34.2  | 44.8  | 63.7  | 77.2  | 85.7  | 89.2  | 88.2  | 75.6  | 60.8  | 46.5  | 37.0  | (25) |
| (26) |                                                                   | MCWB        | 33.2  | 33.3  | 40.6  | 51.4  | 60.1  | 63.8  | 65.3  | 64.2  | 58.4  | 53.0  | 44.5  | 36.1  |       | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 41.4  | 41.4  | 50.1  | 60.5  | 66.9  | 70.6  | 71.0  | 70.8  | 66.7  | 61.8  | 53.2  | 45.0  | (27) |
| (28) |                                                                   | MADB        | 42.8  | 43.8  | 57.3  | 72.9  | 80.5  | 86.9  | 86.8  | 89.1  | 82.2  | 72.4  | 58.1  | 45.5  |       | (28) |
| (29) |                                                                   | 2%          | WB    | 37.9  | 37.8  | 46.2  | 57.5  | 65.1  | 68.9  | 69.4  | 68.7  | 64.2  | 58.5  | 49.9  | 42.0  | (29) |
| (30) |                                                                   | MADB        | 38.7  | 39.2  | 51.7  | 68.9  | 79.3  | 85.4  | 85.1  | 85.9  | 75.8  | 65.4  | 52.0  | 42.8  |       | (30) |
| (31) |                                                                   | 5%          | WB    | 34.8  | 35.4  | 43.1  | 55.3  | 63.1  | 67.3  | 68.2  | 67.2  | 62.2  | 56.2  | 47.1  | 39.1  | (31) |
| (32) |                                                                   | MADB        | 35.2  | 36.4  | 47.8  | 64.5  | 77.2  | 83.2  | 84.6  | 84.2  | 73.9  | 62.4  | 48.8  | 39.9  |       | (32) |
| (33) |                                                                   | 10%         | WB    | 33.0  | 33.5  | 40.5  | 52.8  | 61.2  | 65.6  | 67.0  | 65.8  | 60.1  | 53.7  | 44.3  | 35.8  | (33) |
| (34) |                                                                   | MADB        | 33.4  | 34.1  | 44.4  | 61.8  | 73.8  | 80.4  | 83.5  | 83.1  | 71.9  | 59.7  | 46.2  | 36.4  |       | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 10.0  | 11.4  | 13.5  | 19.0  | 20.6  | 21.0  | 21.1  | 22.1  | 19.8  | 15.8  | 10.7  | 9.5   | (35)  |      |
| (36) |                                                                   | 5% DB       | MADB  | 8.6   | 9.8   | 20.5  | 23.9  | 25.2  | 24.4  | 24.5  | 25.3  | 24.1  | 21.5  | 14.3  | 9.6   | (36) |
| (37) |                                                                   | MCWBR       | 8.0   | 8.3   | 13.2  | 11.9  | 10.7  | 8.6   | 8.3   | 8.5   | 9.8   | 11.4  | 10.9  | 8.8   | (37)  |      |
| (38) |                                                                   | 5% WB       | MADB  | 8.2   | 9.2   | 18.2  | 21.3  | 22.1  | 21.7  | 21.3  | 22.2  | 20.7  | 18.5  | 12.6  | 9.6   | (38) |
| (39) |                                                                   | MCWBR       | 7.9   | 8.2   | 12.5  | 11.8  | 10.6  | 8.9   | 8.2   | 8.9   | 10.6  | 11.3  | 10.7  | 9.2   | (39)  |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.328 | 0.360 | 0.388 | 0.446 | 0.414 | 0.405 | 0.421 | 0.430 | 0.395 | 0.365 | 0.341 | 0.329 | (40)  |      |
| (41) |                                                                   | taud        | 2.131 | 2.040 | 2.134 | 2.066 | 2.248 | 2.317 | 2.273 | 2.227 | 2.293 | 2.327 | 2.283 | 2.178 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 219   | 239   | 257   | 254   | 269   | 272   | 265   | 256   | 253   | 240   | 217   | 201   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 27    | 37    | 40    | 48    | 42    | 39    | 40    | 41    | 34    | 28    | 23    | 23    | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 288   | 582   | 982   | 1445  | 1904  | 2074  | 2017  | 1787  | 1283  | 735   | 339   | 203   | (44)  |      |
| (45) |                                                                   | RadStd      | 59    | 113   | 123   | 109   | 159   | 181   | 124   | 122   | 105   | 98    | 60    | 37    | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | +0.90 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +220  | N/A   |
| (47) | Regional (5 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +220  | N/A   |

Nomenclature: See separate page